Suvi Häkkinen, PhD

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EDUCATION

Ph.D. in Psychology, University of Helsinki, Finland 8/2018
Thesis: "Functional organization of human auditory cortex during active auditory tasks"

M.Sc. (Tech.) in Bioinformation Technology, Aalto University, Finland 10/2012
Major: Computational and Cognitive Biosciences
Minor: Acoustics and Audio Signal Processing
Thesis: "Temporal dynamics of task-dependent activations of human auditory cortex: an EEG study"

B.Sc. (Tech.) in Bioinformation Technology, Aalto University, Finland 8/2012
Major: Computational and Cognitive Biosciences
Minor: Usability in Software Development
Thesis: "Intensity-based registration methods of magnetic resonance images"

RESEARCH EXPERIENCE

Assistant Project Scientist 9/2024 – Present
Postdoctoral Scholar 9/2023 – 9/2024
Helen Wills Neuroscience Institute, University of California, Berkeley
Feinberg Lab (PI David Feinberg)

- Scientific lead for a high-resolution hippocampal imaging project on the NexGen 7T; established that CBV VASO based mapping provides superior depth-dependent localization by mitigating pial-vessel bias and venous contamination common in traditional BOLD-fMRI.
- Evaluated the impact of pulse sequence modifications on point-spread functions and activation patterns, providing empirical evidence to optimize pulse sequences for mesoscale imaging.

Postdoctoral Scholar 1/2022 – 9/2023
Helen Wills Neuroscience Institute, University of California, Berkeley
Cognitive Neuroanatomy Lab (PI Kevin Weiner) & Building Blocks of Cognition Laboratory (PI Silvia Bunge)

- Spearheaded a precision neuroanatomy project (published in *J. Neurosci*) investigating the functional relevance of cortical folding using a manually labeled dataset of 1,800+ sulci.
- Developed a multivariate analytical framework using SVM classification and clustering to demonstrate that individual sulcal morphology provides is a promising "coordinate space".
- Utilized graph theoretic analyses to link tertiary sulcal depth with network centrality, identifying a novel relationship between cortical folding and functional hierarchy.

Postdoctoral Scholar 8/2018 – 12/2021
Memory and Aging Center, University of California, San Francisco (parental leave 5 mo)
Dementia Imaging Genetics Lab (PI Suzee Lee)

- Managed a project integrating data from 17 international sites to characterize neurodegenerative trajectories in presymptomatic and symptomatic *C9orf72*, *GRN*, and *MAPT* mutation carriers.
- Modeled lifespan trajectories of regional volumes and functional networks; identified putative early biomarkers in the presymptomatic stage using linear mixed-effects and normative-like modeling.
- Awarded a competitive personal research fellowship, securing two years of independent funding.

Researcher 1 – 7/2018
University of Turku, Finland
FinnBrain Birth Cohort Study (PI Hasse Karlsson)

- Engineered robust fMRI preprocessing and functional connectivity pipelines optimized for neonatal imaging; used in 6 publications and 4 conference abstracts to date.

Graduate Researcher 9/2013 – 1/2018

Faculty of Medicine, University of Helsinki, Finland (parental leave 10 mo**)**

@brain research group (PI Teemu Rinne)

- Headed five investigations into the functional architecture of human auditory cortex and the effects of different cognitive tasks; results published in 3 peer-reviewed journals.
- Established comprehensive analytical pipelines for fMRI, fMRS (choline-related plasticity), DWI (tractography), and EEG/MEG (source localization) to map task-specific attentional effects.
- Independently performed 100+ MRI sessions, maintaining full responsibility for subject safety, protocol troubleshooting, and data acquisition.

Research Assistant 3/2008 – 12/2010

Institute of Psychology, University of Helsinki, Finland (part time)

Attention and Memory Networks research group (PIs Kimmo Alho and Teemu Rinne)

- Gained proficiency in 3T MRI operation (Siemens and GE platforms), supporting high-volume data collection for audiovisual research.
- Developed a custom Python tool for generating flattened cortical patches, enabling 2D visualization of complex sensory area morphology.

SKILLS

Technical & Analytical Expertise

Imaging Modalities: fMRI (BOLD, VASO, GRASE) • DWI • fMRS • EEG/MEG. **Analytical Frameworks:** GLM/LME analysis • multivariate modeling • non-parametric permutation (spatial null models/spin testing) • machine learning. **Network Science:** connectivity (seed, PPI, local measures) • community detection • graph theory. **High-Resolution Mapping:** layer-fMRI • hippocampal & auditory subfields • cortical morphology. **Diverse Populations:** neurodegenerative cohorts • neonatal & pediatric MRI.

Software, Programming & Infrastructure

Neuroimaging Software: FreeSurfer • LayNII • FSL • AFNI • SPM • Neuropointillist • fMRIPrep • MNE • EEGLAB. **Programming & Data:** Python (NumPy, SciPy, pandas, sklearn, TensorFlow) • MATLAB • R • C/C++ • shell scripting • SQL. **Computing & Infrastructure:** HPC management (SGE/Slurm) • containerized pipelines (Docker/Singularity) • Git/GitHub • BIDS standard • Jupyter • LaTeX. **Experimental tools:** PsychoPy • Presentation.

Languages & Professional Development

Languages: Finnish (Native) • English (full professional) • Swedish • German • French • Spanish. **Training:** Coursera/DeepLearning.AI: Neural Networks, Optimization, and ML Strategy.

PUBLICATIONS (* denotes equal contribution/co-first authorship)

Peer-reviewed publications

Vartiainen, E., Copeland, A. Pulli, E., Kumpulainen V., Silver, E. Rajasilta, O., Jolly, A., Luotonen, S., Kholis Audah, H., Bano, W., Suuronen, I., Saukko, E., **Häkkinen (née Talja), S.**, Karlsson, H. , Karlsson, L., Tuulari, J.J. (2025) Pre -and postnatal maternal depressive symptoms associate with local connectivity of the left amygdala in 5-year-olds. *European Psychiatry*, doi:10.1192/j.eurpsy.2025.10097

Flagan, T. M., Chu, S.A., **Häkkinen, S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2025) Functional connectivity associations with markers of disease progression in *GRN* mutation carriers. *Annals of Clinical and Translational Neurology,* 12(12):2575–2588.

Häkkinen, S., Voorhies, I.W., Willbrand, E.H., Yi-Heng, T., Gagnant, T., Yao, J.K., Weiner, K.S., Bunge, S.A. (2025) Anchoring functional connectivity to individual sulcal morphology yields insights in a pediatric study of reasoning. *Journal of Neuroscience,* 45(26):e0726242025.

Park, S., Beckett, A., **Häkkinen, S.,** Ma, S.J., Kim, H., Feinberg, D.A. (2025) Higher spatial resolution and sensitivity in whole brain functional MRI at 7T using 3D EPI accelerated with variable density CAIPI sampling and temporal random walk. *Magnetic Resonance in Medicine*, 94(2), 678–693.

Zhang, L., Flagan, T.M., **Häkkinen, S.**, Chu, S.A., Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2023) Network connectivity alterations in presymptomatic and symptomatic MAPT mutation carriers. *Annals of Neurology*, 94(4), 632–646.

Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., ... Tuulari, J.J. (2023) Maternal psychological distress associates with alterations in resting-state low-frequency fluctuations and distal functional connectivity of the neonate medial prefrontal cortex, *European Journal of Neuroscience*, 57, 242–257.

Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. (2022) Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Frontiers in Neuroscience*, 16.

Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., Lehtola, S. J., ... Tuulari, J.J. (2021) Increased maternal BMI during pregnancy is associated with altered neonate brain local synchrony in the left superior frontal gyrus, *Scientific Reports*, 11, 19182.

Häkkinen, S., Chu, S.A., Lee, S.E. (2020) Neuroimaging in genetic frontotemporal dementia and amyotrophic lateral sclerosis. *Neurobiology of Disease*, 145, 205063.

Rajasilta, O., Tuulari, J.J., Björnsdotter, M., Lehtola, S. J., Saunavaara, J., **Häkkinen, S.**, Merisaari, H., Parkkola, R., Lähdesmäki, T., Karlsson, L., Karlsson, H. (2020) Resting state networks of the neonate brain identified in independent component analysis. *Developmental Neurobiology*, 80, 111–125.

Häkkinen, S., Rinne, T. (2018) Intrinsic, stimulus-driven and task-dependent connectivity in human auditory cortex. *Brain Structure & Function*, 223(5), 2113–2127.

Häkkinen, S., Ovaska, N., Rinne, T. (2015) Processing of pitch and location in human auditory cortex during visual and auditory tasks. *Frontiers in Psychology*, 6.

Talja, S., Alho, K., Rinne, T. (2015) Source analysis of event-related potentials during pitch discrimination and pitch memory tasks. *Brain Topography*, 28, 445–458.

Rinne, T., Koistinen, S., **Talja, S.**, Wikman, P., Salonen, O. (2012) Task-dependent activations of human auditory cortex during spatial discrimination and spatial memory tasks. *NeuroImage*, 59, 4126–4131.

Works currently under review

Häkkinen, S., Beckett, A., Walker, E., Huber, L., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T, https://doi.org/10.1101/2025.08.29.673151

Zhang, L.*, **Häkkinen, S.***, Chu, S. A., Flagan, T. M., Coppola, G., Geschwind, D. H., ... Lee, S. E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Longitudinal functional network connectivity changes across the clinical stages of *C9orf72* hexanucleotide repeat expansion carriers, https://doi.org/10.1101/2025.11.23.25340528

Copeland, A., Rajasilta, O., **Häkkinen, S.**, Kumpulainen, V., Silver, E., Pulli, E.P., Saukko, E., Jolly, A., Saunavaara, J., Parkkola, R., Lähdesmäki, T., Korja, R., Karlsson, L., Karlsson, H., Tuulari, J.J. Functional network organization of the 5-year-old brain identified using independent component analysis and evaluation of denoising strategies, submitted.

CONFERENCE ABSTRACTS

Ru, H., **Häkkinen, S.**, Beckett, A., Walker, E., Feinberg, D.A., Park, S. Accelerated 3D Zoomed GRASE for Mesoscale fMRI using Self-Supervised Reconstruction (DeepGRASE) [Abstract submitted for presentation]. *OHBM,* 2026.

Beckett, A., Walker, E.B., **Häkkinen, S.**, Khegai, O., Vu, A., Huber, R., Feinberg, D. Studying cortical networks using whole brain cerebral blood volume weighted imaging on the NexGen 7T [Abstract submitted for presentation]. *OHBM*, 2026.

Beckett, A.J., **Häkkinen, S.**, Walker, E.B., Khegai, O., Vu, A., Huber, R., & Feinberg, D. (2026, May). Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner [Abstract submitted for presentation]. *ISMRM*, 2026.

Häkkinen, S., Voorhies, W.I., Willbrand. E.H.,Tsai, Y.-H., Yao. J.K., Weiner, K.S.,, Bunge, S.A. Lateral frontoparietal functional connectivity based on individual sulcal morphology. *SfN*, 2025.

Walker, E. B, Beckett, A. J., **Häkkinen, S.**, Vu, A., Huber, R., & Feinberg, D. Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner. *SfN*, 2025

Zhang, L., **Häkkinen**, **S.**, Jung, Y., Mandelli, M.L., Leichter, D., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia.. *AAIC*, 2025. Longitudinal functional connectivity changes across the clinical spectrum of *C9orf72* hexanucleotide repeat expansion carriers. *AAIC*, 2025.

Häkkinen, S., Beckett, A., Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM,* 2025.

Beckett, A.*, **Häkkinen, S.***, Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM* Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, 2025. (**Best Poster Award**)

Beckett, A., Park, S., **Häkkinen, S.**, Walker, E., Vu, A., Feinberg, D.A. Development of GRASE Pulse Sequence with larger field of view for mesoscale functional MRI on the Next Generation (NexGen) 7T scanner. *ISMRM,* 2025.

Ma, S.J., Walker, E., **Häkkinen, S.**, Beckett, A.J.S., Vu, A.T., Feinberg, D.A. NexGen 7T MRI scanner for mesoscale brain imaging. *ANA,* 2024.

Flagan, T.M., Chu, S.A., **Häkkinen, S**., Zhang, L., ... Seeley. W.W., Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Functional connectivity associations with markers of disease progression in *GRN* mutation carriers. *AAIC*, 2024.

Huber, R., Beckett, A., Ma, S., Stirnberg, R., Morgan, T., Ehses, P., **Häkkinen, S.,** Gunamony, S., Bandettini, P., Feinberg, S. Developing whole brain layer-fMRI protocols on the NexGen 7T scanner. *OHBM,* 2023.

Beckett, A.J.S., Huber, R., Ma, S. J., **Häkkinen, S.,** Gunamony, S., Feinberg, D.A. Whole brain layer-fMRI on the NexGen 7T scanner with high performance gradients and 64-channel receiver array. *ISMRM,* 2023.

Park, S., Beckett, A.J.S., **Häkkinen, S.,** Ma, S. J., Feinberg, D.A. Whole-brain sub-millimeter resolution fMRI with 3D EPI accelerated with temporal random walk. *ISMRM,* 2023.

Lombardi, J., Zhang, L., Flagan, T.M., Vargas, A.G., Mandelli, M.L., Brown, J.A., **Häkkinen, S.**, ... Miller, B.L., Gorno-Tempini, M.L., Seeley, W.W., Lee, S.E. on behalf of the ALLFTD Consortium. Structural and functional alterations in young adult presymptomatic frontotemporal lobar degeneration mutation carriers. *AAIC*, 2023.

Zhang, L., Flagan, T.M., Staffaroni, A., **Häkkinen, S.**, Brown, J.A., Mandelli, M.L., ... Gorno-Tempini, M.L., Miller, B.L., Seeley, W.W., Lee, S.E. on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Functional connectivity trajectories in genetic frontotemporal dementia. *AAIC*, 2023.

Lombardi., J., Zhang, L., Flagan, T. M., Mandelli, M. L., Brown, J. A., **Häkkinen, S.**, ... Lee, S.E., on behalf of the ALLFTD Consortium. Salience network alterations in young adult presymptomatic FTLD mutation carriers. *ISFTD,* 2022.

Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Flux Congress* 2022.

Flagan T. M., Chu S.A., **Häkkinen S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Complement and NfL associations with brain structure and functional connectivity alterations in presymptomatic and symptomatic *GRN* mutation carriers. *AAIC 2021*.

Zhang, L., Flagan, T.M., Chu, S.A., **Häkkinen, S.**, Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Presymptomatic and symptomatic *MAPT* mutation carriers feature functional connectivity alterations. *AAIC,* 2021.

Häkkinen, S., Chu, S.A., Flagan, T. M., Gendron, T. D., Petrucelli, L., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Associations of poly(GP), NfL and functional network alterations in *C9orf72* expansion carriers. *OHBM,* 2020.

Zhang, L., Flagan, T. M., Chu, S.A., **Häkkinen, S.**, Rojas-Martinez, J., ... Lee, S.E., on behalf of the ALLFTD Consortia. Brain functional connectivity changes in presymptomatic and symptomatic *MAPT* mutation carriers. *OHBM,* 2020.

Tuulari, J.J., **Häkkinen, S.**, Rajasilta, O., Lehtola, S., Merisaari, H., ... Karlsson, H. Maternal prenatal anxiety associates to infant orbitofrontal cortex functional synchrony. *Flux Congress*, 2018.

Rajasilta, O., Tuulari J.J., **Häkkinen, S.**, Lehtola, S., Merisaari, H., ... Karlsson, H. Functional networks in term-born infants – a resting-state functional MRI study from FinnBrain Birth Cohort study. *Flux Congress,* 2018.

Rinne, T., **Talja, S.**, Stecker, G.C. (2015) Processing in auditory cortex depends on the characteristics of the discrimination task: fMRI study. *The 38th MidWinter Meeting of The Association for Research in Otolaryngology,* 2015.

Stecker, G.C., Higgins, N., McLaughlin, S., **Talja, S.**, Rinne, T. (2015) Stimulus- and task-dependent spatial processing in the human auditory cortex. *The 38th MidWinter Meeting of The Association for Research in Otolaryngology,* 2015.

Talja, S., Ovaska, N., Rinne, T. (2014) Feature-specific and task-dependent processing of pitch and location in human auditory cortex. *OHBM,* 2014.

Rinne, T., Ovaska, N., **Talja, S.** (2013) Task-dependent activations of human auditory cortex during pitch and spatial tasks. *The 36th MidWinter Meeting of The Association for Research in Otolaryngology*, 2013.

DATASETS

Häkkinen, S., Beckett, A., Walker, E., Huber, R., Feinberg, D. (2025) 7T fMRI VASO of the human hippocampus. OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds007122.v1.0.0

GRANTS AND FELLOWSHIPS

Päivi and Sakari Sohlberg Foundation (Sep 2019, 140 000 €), to support my postdoctoral research on genetic dementia biomarkers

Instrumentarium Science Foundation (Jan 2016, 18 000 €), to support my doctoral research in auditory neuroscience

University of Helsinki Doctoral Fellowship (2014 – 2016), Doctoral Programme of Psychology, Learning and Communication, University of Helsinki, Finland

MENTORSHIP & EDUCATIONAL OUTREACH

Research Mentor, Feinberg, Weiner and Bunge Labs, UC Berkeley 2022 – Present

- Guided graduate students and research assistants in MRI analysis, experimental design, and scientific writing

Discussion Lead, Cognitive Neuroscience Colloquium, UC Berkeley 2022

- "Brain-wide association studies: Challenges and opportunities"

Speaker, MAC Postdoc Educational Curriculum, Memory and Aging Center, UCSF 2019

- "NMR basics and MRI contrasts"

Research Mentor, University of Helsinki 2011 – 2013

- Mentored a student through the successful completion of a Master's thesis
- Tutored interns in subject recruitment and behavioral training

PROFESSIONAL RECOGNITION & SERVICE

Invited speaker, layer-fMRI talk (webinar for NIH/Maastricht/MGH researchers) 3/2026

Best Poster Award (Joint 1st Author): ISMRM Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, 2025.

Ad-Hoc Reviewer: *Human Brain Mapping*, *Developmental Cognitive Neuroscience*

Public Outreach: Research on tertiary sulci featured in *UC Berkeley News*

PROFESSIONAL MEMBERSHIPS

Society of Neuroscience (SfN) 2025

The International Society for Magnetic Resonance in Medicine (ISMRM) 2025

Organization for Human Brain Mapping (OHBM) 2014, 2020